

ST05

Food and Nutrition

Science and Technology carries 18-23 questions in every RAS paper. Food and nutrition reliably supplies 2-4 of them, and the vitamin grid alone is worth a question in most papers. The nutrition schemes get counted again in the Current Affairs and Rajasthan blocks, so the return per hour here is among the highest in the syllabus.

This chapter is pure marks. Food and nutrition is the one biology topic where RPSC does not ask you to understand a mechanism - it asks you to match a name to a name. Vitamin to chemical name, vitamin to deficiency disease, adulterant to food item, scheme to launch year. Learn the tables cold and you convert every question in this chapter.

Food and Nutrition

Balanced diet

- Carbs 55-60%, fat 20-30%, protein 10-15%
- Carb 4, protein 4, fat 9, alcohol 7 kcal/g
- Five food groups plus water
- Water = 60-70% of body weight

Vitamins (biggest)

- Fat-soluble: A D E K
- Water-soluble: B-complex and C
- A - night blindness; D - rickets
- B1 beriberi, B3 pellagra, C scurvy
- B12 contains cobalt

Minerals

- Iodine - goitre, cretinism
- Iron - anaemia
- Calcium - rickets, osteoporosis, tetany
- Fluoride - less gives caries, more gives fluorosis

Malnutrition

- Kwashiorkor = protein only, oedema
- Marasmus = protein + calorie, wasting
- Stunting / wasting / underweight
- BMI = kg divided by m squared

Better food

- Fortification = added in processing, +F logo
- Biofortification = bred into the crop
- Dhanashakti bajra, Golden Rice, QPM maize

Food safety

- LTLT, HTST, UHT, irradiation
- Argemone, kesari dal, metanil yellow
- FSSAI 2006, Health Ministry, New Delhi
- Eat Right India, RUCO, trans fat 2%

Programmes

- ICDS 1975
 - NFSA 2013
 - POSHAN Abhiyaan 2018 - Jhunjhunu
 - PM POSHAN 2021
 - Anaemia Mukh Bharat
- 6x6x6

Balanced diet - what the phrase actually means

A balanced diet is not a diet with plenty of everything. It is a diet in which each nutrient comes in the right proportion to the others, in the quantity the body actually needs. Two things must balance at once - the total energy, and the share of that energy contributed by each macronutrient. Get the energy

right but the shares wrong and you still get disease. Food groups are the shortcut: take something from each group and the proportions look after themselves.

- Standard energy split - carbohydrate 55-60%, fat 20-30%, protein 10-15% of total calories.
- Water makes up 60-70% of body weight, the single largest component of the body.
- Nutrients split into macronutrients (carbohydrate, protein, fat) and micronutrients (vitamins, minerals).
- Dietary fibre gives almost no calories but keeps the bowel working - cellulose, hemicellulose, pectin, lignin, gums.

Five food groups and what each one is really for

Group	Examples	Chiefly supplies
Cereals, millets, pulses	Wheat, rice, bajra, ragi, gram	Energy plus plant protein
Vegetables and fruits	Green leafy, roots, citrus	Vitamins, minerals, fibre
Milk and animal foods	Milk, curd, egg, fish, meat	Good-quality protein, calcium, B12
Fats, oils and nuts	Ghee, oil, groundnut	Concentrated energy, fat-soluble vitamins
Sugars	Sugar, jaggery, honey	Energy only, nothing else

TRAP

Dietary fibre is plant material your enzymes cannot digest. Glycogen is animal starch and IS digestible - so glycogen is NOT dietary fibre. RPSC sets this as a 'which one is not a constituent of dietary fibre' question.

Macronutrients and their calorific values

Calorific value is the energy released when one gram of a nutrient is completely oxidised in the body. There are only four numbers in the whole list, and the one students skip - alcohol - is the one the paper likes. Learn all four, in order.

Calorific values - four numbers, learn them cold

Nutrient	Energy per gram	Note
Carbohydrate	4 kcal	The body's first-choice fuel
Protein	4 kcal	Same as carbohydrate - a favourite trap
Fat	9 kcal	Highest, more than double carbohydrate
Alcohol	7 kcal	Second highest, supplies no other nutrient

- Proteins are chains of amino acids; nine are essential for adult humans and must come from food.
- The nine - histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, valine.
- Egg protein is the reference protein, the standard other proteins are scored against.
- Cereal protein is low in lysine, pulse protein is low in methionine - eaten together they complete each other, which is why dal-chawal works.
- Protein is for growth and repair; carbohydrate and fat are for energy. Burning protein for fuel is a waste.

YAAD RAKHO

Descending order of calorific value - Fat 9, Alcohol 7, Carbohydrate 4, Protein 4. Say it as nine-seven-four-four.

Vitamins – the single most examined table in this chapter

Vitamins are organic compounds needed in tiny amounts that the body cannot make in sufficient quantity. The paper picks one of four angles every time – the chemical name, the source, the deficiency disease, or fat-soluble versus water-soluble. So learn all four columns together, never the disease alone. And learn the solubility split first, because that one line answers a whole question by itself.

- Fat-soluble – A, D, E, K. They dissolve in fat, are stored in liver and fat tissue, and an excess can therefore poison you: hypervitaminosis.
- Water-soluble – the whole B-complex and vitamin C. Not stored, the excess leaves in urine, so they must be eaten regularly.
- Because the fat-soluble ones are stored, their deficiency takes months to appear. Water-soluble deficiency shows up quickly.

YAAD RAKHO

Fat-soluble vitamins spell A-DEK. Everything else – all the B's and C – is water-soluble. If tocopherol (vitamin E) turns up in a list of water-soluble vitamins, that is your odd one out.

Fat-soluble vitamins – chemical name, source, deficiency

Vitamin	Chemical name	Good sources	Deficiency
A	Retinol	Carrot, green leafy vegetables, liver, milk	Night blindness, Bitot's spots, xerophthalmia, keratomalacia
D	Calciferol – D2 ergocalciferol, D3 cholecalciferol	Sunlight on skin, fish liver oil, egg yolk	Rickets in children, osteomalacia in adults

Vitamin	Chemical name	Good sources	Deficiency
E	Tocopherol	Vegetable oils, wheat germ, nuts	Antioxidant; deficiency causes sterility in rats
K	Phylloquinone K1, menaquinone K2	Green leafy vegetables; K2 made by gut bacteria	Delayed blood clotting, excess bleeding

- Vitamin D is the one your own skin makes - from 7-dehydrocholesterol, using UV-B in sunlight. Hence 'sunshine vitamin'.
- Vitamin K is needed to synthesise prothrombin, the clotting protein - which is why newborns are given a vitamin K shot.
- Bitot's spots are greyish triangular patches on the white of the eye, a clinical sign of vitamin A deficiency.

Water-soluble vitamins - B-complex and C

Vitamin	Chemical name	Deficiency disease
B1	Thiamine	Beriberi
B2	Riboflavin	Cheilosis, angular stomatitis
B3	Niacin / nicotinic acid	Pellagra - dermatitis, diarrhoea, dementia
B5	Pantothenic acid	Burning feet syndrome
B6	Pyridoxine	Anaemia and convulsions
B7	Biotin	Dermatitis and hair loss
B9	Folic acid	Megaloblastic anaemia; neural tube defects in the foetus
B12	Cyanocobalamin	Pernicious anaemia
C	Ascorbic acid	Scurvy - bleeding gums, delayed wound healing

- Vitamin B12 is the vitamin that contains a metal - cobalt. That single fact is a question on its own.
- B12 comes only from animal sources and microbial fermentation, so strict vegetarians are the risk group.

- Folic acid is given in pregnancy to prevent neural tube defects - the reason it goes into fortified wheat flour.
- Humans, other primates and guinea pigs cannot synthesise vitamin C; most other animals can.

ASKED IN RAS

Vitamin-to-chemical-name and vitamin-to-deficiency matching is a standing feature of the RPSC science block - the question bank for this chapter alone carries four separate matching sets built on it. The four pairs that recur: B1 thiamine - beriberi, B3 niacin - pellagra, B12 cyanocobalamin - pernicious anaemia, C ascorbic acid - scurvy.

TRAP

Pyridoxine (B6) causes anaemia and convulsions, NOT pellagra - pellagra is niacin (B3). The 'not correctly matched' question is built on exactly this swap. Second common swap: riboflavin paired with beriberi, when beriberi belongs to thiamine.

Minerals - fewer names, sharper numbers

Minerals are inorganic, so unlike vitamins they cannot be destroyed by cooking - they only leach into the cooking water. Two angles get asked: which deficiency causes which disease, and the two policy numbers, iodine in salt and fluoride in water. Fluoride is the tricky one, because both too little and too much cause disease.

Minerals and their deficiency effects

Mineral	Main role	Deficiency
Iodine (I)	Thyroid hormone synthesis	Goitre in adults, cretinism in children

Mineral	Main role	Deficiency
Iron (Fe)	Haemoglobin	Anaemia
Calcium (Ca)	Bone, teeth, clotting, muscle contraction	Rickets, osteoporosis, tetany
Zinc (Zn)	Enzymes, immunity, growth	Growth retardation, poor wound healing
Fluoride (F)	Tooth enamel	Too little - dental caries; too much - dental and skeletal fluorosis
Sodium (Na), Potassium (K)	Fluid balance, nerve impulse	Cramps and weakness; excess sodium raises blood pressure

- Under NIDDCP, iodised salt must carry not less than 30 ppm iodine at manufacture and 15 ppm at the consumer end - the gap allows for losses in transit.
- BIS drinking-water standard for fluoride - desirable limit 1.0 mg/L, permissible limit 1.5 mg/L. Rajasthan has serious fluorosis belts, so this is a state-relevant number.
- Iodine deficiency in pregnancy causes cretinism, stunted growth with mental retardation; goitre is the adult sign.
- Double Fortified Salt (DFS) carries both iodine and iron.

ASKED IN RAS

Endemic goitre as a direct consequence of dietary iodine deficiency is a recurring one-liner, and its companion is the 30 ppm / 15 ppm iodised-salt standard. Learn the disease and the number together - the paper has asked both faces.

Malnutrition - PEM, the three indices, anaemia and BMI

Malnutrition does not mean only 'too little food'. It means the wrong balance - and the two classic protein-energy malnutrition conditions differ on exactly that balance. Kwashiorkor is the child getting calories but not protein. Marasmus is the child getting neither. Learn them as a pair, because that is how the question is set.

Kwashiorkor vs Marasmus

Kwashiorkor	Marasmus
Protein deficiency with nearly adequate calories	Protein AND calorie deficiency together
Oedema present - swollen legs and belly	No oedema
Moon face, flaky-paint dermatosis	Extreme wasting, 'old man's face'
Usually after weaning, about 1-3 years	Usually below one year of age
Some subcutaneous fat retained	Fat and muscle both lost
Term coined by Cicely Williams from a Ghanaian word	Name from the Greek for wasting

- Stunting = low height-for-age. It reflects chronic, long-term undernutrition.
- Wasting = low weight-for-height. It reflects acute, recent undernutrition.
- Underweight = low weight-for-age, a composite of the other two.
- NFHS-5 (2019-21) India - stunting 35.5%, wasting 19.3%, underweight 32.1%.
- NFHS-5 anaemia - 67.1% of children aged 6-59 months and 57.0% of women aged 15-49.

Anaemia - WHO haemoglobin cut-offs

Group	Anaemia if haemoglobin below
Adult men	13 g/dL
Non-pregnant women	12 g/dL
Pregnant women	11 g/dL
Children 6-59 months	11 g/dL

- BMI = weight in kilograms divided by the square of height in metres; the unit is kg/m².
- WHO bands - below 18.5 underweight, 18.5-24.9 normal, 25-29.9 overweight, 30 and above obese.
- Asian-Indian cut-offs are lower - 23 and above overweight, 25 and above obese - because Indians carry more body fat at the same BMI.
- India now carries a double burden - undernutrition and obesity together in the same population.

ASKED IN RAS

Kwashiorkor as protein deficiency with oedema is a repeatedly examined statement set. The statement that trips people is the age - kwashiorkor is the post-weaning child, marasmus the infant under one year.

Fortification and biofortification - know the difference

Both raise the nutrient content of food, but at completely different points in the chain. Fortification happens after harvest, in the factory - you add the nutrient to the food, and FSSAI marks such packs with the '+F' logo.

Biofortification happens before harvest, in the field - you breed or grow the nutrient into the crop itself. One word gives it away: 'bio' means it was grown that way - fortification is Factory, biofortification is Bred in the field. Fortified rice now moves through the PDS, ICDS and PM POSHAN. Watch the odd one out in the biofortified list: Golden Rice gets its beta-carotene by genetic engineering, while Dhanashakti and QPM come from ordinary breeding.

Fortification - which staple carries which nutrient

Food vehicle	Nutrient added
Salt	Iodine; iron as well in Double Fortified Salt

Food vehicle	Nutrient added
Wheat flour and rice	Iron, folic acid, vitamin B12
Milk and edible oil	Vitamins A and D

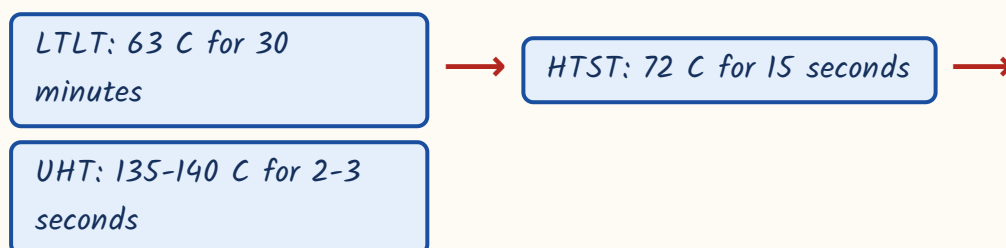
Biofortified varieties worth naming

Variety / crop	Nutrient	Note
Dhanashakti - pearl millet (bajra)	Iron	India's first notified biofortified variety
Golden Rice	Provitamin A (beta-carotene)	Genetically engineered, not conventionally bred
QPM - Quality Protein Maize	Lysine and tryptophan	Fixes the amino-acid gap of ordinary maize

Food preservation, adulteration and FSSAI

Food spoils because microbes and enzymes work on it. Every preservation method therefore does one of three things - kill the microbes, take away the water they need, or change the conditions so they cannot grow. Once you see the list that way it stops being arbitrary. Pasteurisation is the one asked by its numbers, so learn the temperature-time ladder: as the temperature rises, the time falls.

Pasteurisation - hotter means shorter



Preservation methods and the names attached to them

Method	How it works	Name to remember
Canning	Sealed, then heat sterilised	Invented by Nicolas Appert
Irradiation	Cobalt-60 gamma rays kill microbes and stop sprouting	Carries the Radura logo
Lyophilisation	Freeze drying - water removed by sublimation	Instant coffee, vaccines
Dehydration, salting, sugaring	Removes or locks up water	Papad, pickle, murabba
Refrigeration and freezing	Slows microbial growth	Slows spoilage, does not sterilise

- Class I preservatives are natural - salt, sugar, vinegar, spices, honey, edible oil. No legal limit on their use.
- Class II preservatives are chemical - sodium benzoate, sulphur dioxide, sorbic acid, nitrites. These are limited by law.
- Pasteurisation kills pathogens and cuts the microbial load; it does not sterilise, which is why pasteurised milk still needs a fridge.

Adulteration means adding a cheaper or harmful substance to food. RPSC asks it as a matching set and often as a 'not correctly matched' question, so learn the pairs in both directions - food to adulterant, and adulterant back to the disease it causes.

Food - adulterant - effect

Food	Common adulterant	Effect
Mustard oil	Argemone oil	Epidemic dropsy - oedema of the legs, cardiac complications
Arhar and other dals	Kesari dal (<i>Lathyrus sativus</i>), toxin BOAA/ODAP	Lathyrism - crippling paralysis of the lower limbs
Turmeric powder	Metanil yellow, lead chromate	Liver damage, carcinogenic

Food	Common adulterant	Effect
Black pepper	Papaya seeds	Cheating, gastric irritation
Chilli powder	Brick powder	Grit, gastric problems
Milk	Water, starch, detergent, urea	Dilution, gastrointestinal illness

- FSSAI was set up under the Food Safety and Standards Act, 2006, works under the Ministry of Health and Family Welfare and is headquartered in New Delhi.
- The 2006 Act consolidated the earlier laws, chiefly the Prevention of Food Adulteration Act, 1954.
- An FSSAI licence number has 14 digits and must be printed on the pack.
- FSSAI initiatives - Eat Right India (2018), Eat Right Challenge, RUCO (Repurpose Used Cooking Oil), Clean Street Food Hub, BHOG for places of worship.
- Industrial trans fatty acids in foods have been capped at 2% by weight since January 2022.

TRAP

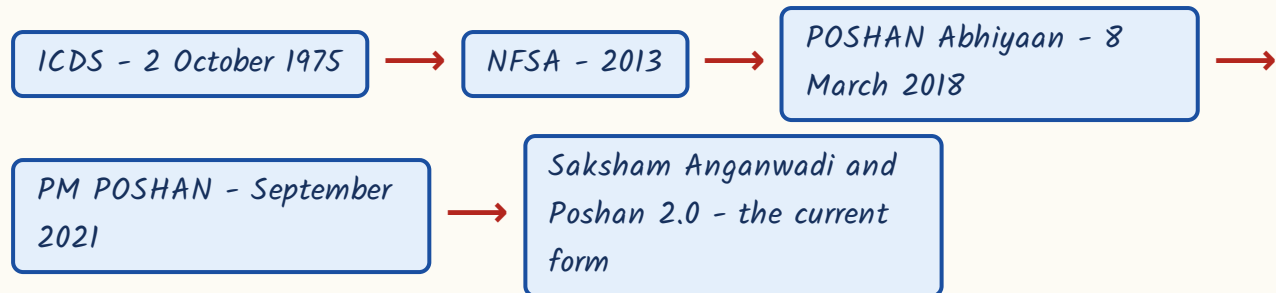
Brick powder adulterates chilli powder, not milk. Milk's usual adulterants are water, starch, detergent and urea. This exact swap is how the 'not correctly matched' option is built.

Nutrition programmes - the scheme grid

These schemes are asked in three different blocks of the paper - Science, Current Affairs and Rajasthan Economy - so the return on learning them is triple. What gets asked is the launch year, the target group and the one

distinguishing feature. Fix the chronology first, because ordering questions are set on it.

Chronology - the order has been asked as an ordering question



The five programmes you must be able to name

Programme	Year	Core feature
ICDS	2 October 1975	Six services through the anganwadi; now runs as Saksham Anganwadi and Poshan 2.0
NFSA	2013	Legal right to foodgrain; covers 75% rural and 50% urban population
POSHAN Abhiyaan (National Nutrition Mission)	8 March 2018	Launched from Jhunjhunu, Rajasthan; targets stunting and anaemia
PM POSHAN (Poshan Shakti Nirman)	September 2021	Replaced the Mid-Day Meal Scheme; covers Bal Vatika and Classes I-VIII
Anaemia Mukht Bharat	2018	6x6x6 strategy - six beneficiary groups, six interventions, six institutional mechanisms

- POSHAN Abhiyaan targets - cut stunting and underweight by 2% a year and anaemia by 3% a year.
- NFSA entitlement - 5 kg of foodgrain per person per month for a priority household, 35 kg per household under Antyodaya Anna Yojana.
- NFSA issue prices - rice Rs 3, wheat Rs 2, coarse grain Rs 1 per kg.
- Jhunjhunu as the POSHAN Abhiyaan launch site is the Rajasthan hook RPSC likes. Remember the district name.

ASKED IN RAS

The launch of POSHAN Abhiyaan from Jhunjhunu on 8 March 2018 is exactly the Rajasthan-linked national-scheme fact RPSC turns into a one-liner. Pair it with ICDS 1975 and PM POSHAN 2021 for the chronological-ordering version.

CURRENT LINK

Current link (as of September 2026): 2023 was the UN International Year of Millets, declared on India's proposal, and millets were rebranded 'Shree Anna'. Millets are still the government's headline nutrition-and-climate crop, and Rajasthan is India's largest producer of bajra - so a millet question can reach you through Science, Current Affairs or Rajasthan Agriculture. Ragi is the richest cereal source of calcium.

REVISION RECAP

1. Calorific values - fat 9, alcohol 7, carbohydrate 4, protein 4 kcal per gram.
2. Energy split of a balanced diet - carbohydrate 55-60%, fat 20-30%, protein 10-15%; water is 60-70% of body weight.
3. Fat-soluble A D E K are stored and can cause hypervitaminosis; B-complex and C are water-soluble and are not stored.
4. A retinol - night blindness and Bitot's spots; D calciferol - rickets and osteomalacia; E tocopherol; K phyloquinone - delayed clotting.
5. B1 thiamine - beriberi; B2 riboflavin - cheilosis; B3 niacin - pellagra; B6 pyridoxine - anaemia and convulsions.
6. B9 folic acid - megaloblastic anaemia and neural tube defects; B12 cyanocobalamin - pernicious anaemia, contains cobalt.
7. Vitamin C is ascorbic acid, deficiency scurvy; humans, primates and guinea pigs cannot make it.
8. Iodine - goitre and cretinism; iron - anaemia; calcium - rickets, osteoporosis, tetany; zinc - growth retardation.
9. Iodised salt 30 ppm at manufacture and 15 ppm at consumer end; fluoride in water 1.0 mg/L desirable, 1.5 mg/L permissible.

10. Kwashiorkor - protein only, oedema, post-weaning child. Marasmus - protein plus calories, wasting, infant under one year.
11. Stunting is height-for-age, wasting is weight-for-height, underweight is weight-for-age.
12. BMI is kg per m²; WHO calls 30 obese, the Asian-Indian cut-off is 25.
13. Fortification is done during processing and carries the +F logo; biofortification is bred or grown into the crop.
14. Dhanashakti is India's first biofortified variety, iron-rich bajra; Golden Rice carries beta-carotene; QPM maize carries lysine and tryptophan.
15. Pasteurisation - LTLT 63 C/30 min, HTST 72 C/15 sec, UHT 135-140 C/2-3 sec; irradiation uses Cobalt-60 and the Radura logo.
16. Argemone oil in mustard oil gives epidemic dropsy; kesari dal in arhar gives lathyrism; metanil yellow adulterates turmeric.
17. FSSAI - Act of 2006, Health Ministry, New Delhi, 14-digit licence, trans fat capped at 2% since January 2022.
18. ICDS 1975, NFSA 2013, POSHAN Abhiyaan 8 March 2018 from Jhunjhunu, PM POSHAN 2021 in place of the Mid-Day Meal Scheme.
19. NFSA gives 5 kg per person per month, 35 kg per AAY household, at Rs 3, Rs 2 and Rs 1 per kg.
20. Anaemia Mukt Bharat runs a 6x6x6 strategy; 2023 was the International Year of Millets, branded Shree Anna.